



HACKTHEBOX

Noted Writeup



Prepared by: Cyberjunkie & Sebh24

Machine Author(s): Cyberjunkie

Difficulty: **Easy**

Scenario

Simon, a developer working at forela notified the CERT team about a note appearing on his desktop which claims that his system was compromised and they collected all sensitive data from simon's workstation. They performed data extortion on his workstation and are now threatening to release the data on dark web unless their demands are met. Simon had multiple sensitive files like planned software works, internal development plans and code base of applications. The threat intelligence team believes that the threat actor made some mistakes and the TI team cannot find any way to contact threat actors. The company stakeholders really need this incident to be handled and all the sensitive data back. They are demanding that no matter what, the data should not be leaked.

Since you are our junior security analyst, you are assigned only a specific type of dfir investigation regarding this case. The CERT lead triaged the workstation and provided you only with notepad++ artifacts, since he suspected that attacker created extortion note and other extortion activities with hands-on keyboard access. Now it's your duty to find out how the attack happened and find a way to get in contact with threat actors as they accidentally locked out their information on how to contact them.

Artefacts provided

1- Noted.zip md5 : 528F1181FCE7B59E311F1FAA0D88E316

Skills Learnt

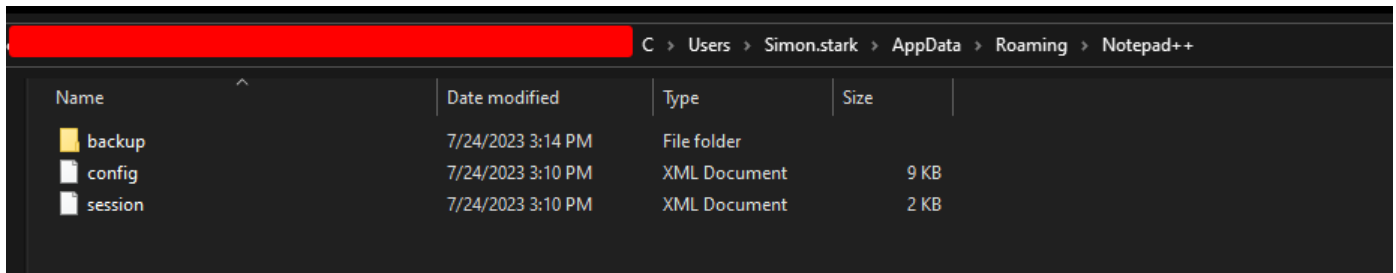
- Windows Forensics
- Data Extortion
- OSINT
- IOC Gathering
- Ransom

Tags

- Windows Forensics
- Notepad ++
- Incident Response

Initial Analysis

We are provided with a KAPE acquisition zip file. KAPE (Kroll Artifact Parser and Extractor) is a powerful tool used for the rapid collection and parsing of forensic artifacts. Upon unzipping and navigating through the artifacts, we find a Notepad++ folder located at `C:\Users\Simon.stark\AppData\Roaming\Notepad++`.



We can see two XML documents and a backup folder. Let's briefly discuss these two XML files.

The `config.xml` file contains configuration settings for the Notepad++ application, such as user preferences and custom settings. It also records a history of files opened using Notepad++. We can open this file in Notepad or any text editor of your choice; no special application is needed.

```
config - Notepad
File Edit Format View Help
<ProjectPanel id="1" workspaceFile="" />
<ProjectPanel id="2" workspaceFile="" />
</ProjectPanels>
<ColumnEditor choice="number">
  <text content="" />
  <number initial="-1" increase="-1" repeat="-1" formatChoice="dec" leadingChoice="none" />
</ColumnEditor>
<GUIConfig>
  <GUIConfig name="ToolBar" visible="yes">standard</GUIConfig>
  <GUIConfig name="StatusBar">show</GUIConfig>
  <GUIConfig name="TabBar" dragAndDrop="yes" drawTopBar="yes" drawInactiveTab="yes" reduce="yes" closeButton="yes" doubleClick2Close="no" vertical="no" multiline="no" hide=
  <GUIConfig name="ScintillaViewsSplitter">vertical</GUIConfig>
  <GUIConfig name="UserDefineDlg" position="undocked">hide</GUIConfig>
  <GUIConfig name="TabSetting" replaceBySpace="no" size="4" />
  <GUIConfig name="AppPosition" x="775" y="6" width="1100" height="700" isMaximized="no" />
  <GUIConfig name="FindWindowPosition" left="0" top="0" right="0" bottom="0" isLessModeOn="no" />
  <GUIConfig name="FinderConfig" wrappedLines="no" purgeBeforeEverySearch="no" showOnlyOneEntryPerFoundLine="yes" />
  <GUIConfig name="noUpdate" intervalDays="15" nextUpdateDate="20230808">no</GUIConfig>
  <GUIConfig name="Auto-detection">yes</GUIConfig>
  <GUIConfig name="CheckHistoryFiles">no</GUIConfig>
  <GUIConfig name="TrayIcon">no</GUIConfig>
  <GUIConfig name="MaintainIndent">yes</GUIConfig>
  <GUIConfig name="TagsMatchHighlight" TagAttrHighlight="yes" HighlightNonHtmlZone="no">yes</GUIConfig>
  <GUIConfig name="RememberLastSession">yes</GUIConfig>
  <GUIConfig name="DetectEncoding">yes</GUIConfig>
  <GUIConfig name="SaveAllConfirm">yes</GUIConfig>
  <GUIConfig name="NewDocDefaultSettings" format="0" encoding="4" lang="0" codepage="-1" openAnsiAsUTF8="yes" addNewDocumentOnStartup="no" />
  <GUIConfig name="langExcluded" gr0="0" gr1="0" gr2="0" gr3="0" gr4="0" gr5="0" gr6="0" gr7="0" gr8="0" gr9="0" gr10="0" gr11="0" gr12="0" langMenuCompact="yes" />
  <GUIConfig name="Print" lineNumber="yes" printOption="3" headerLeft="" headerMiddle="" headerRight="" footerLeft="" footerMiddle="" footerRight="" headerFontName="" header
  <GUIConfig name="Backup" action="0" useCustomDir="no" dir="" isSnapshotMode="yes" snapshotBackupTiming="7000" />
  <GUIConfig name="TaskList">yes</GUIConfig>
  <GUIConfig name="MRU">yes</GUIConfig>
  <GUIConfig name="URL">2</GUIConfig>
  <GUIConfig name="uriCustomizedSchemes">svn:// cvs:// git:// imap:// irc:// irc6:// ircs:// ldap:// ldaps:// news: telnet:// gopher:// ssh:// sftp:// smb:// skype: snmp://
  <GUIConfig name="globalOverride" fg="no" bg="no" font="no" fontSize="no" bold="no" italic="no" underline="no" />
  <GUIConfig name="auto-completion" autoAction="3" triggerFromNbChar="1" autoCignoreNumbers="yes" insertSelectedItemUseENTER="yes" insertSelectedItemUseTAB="yes" autoCbrief
  <GUIConfig name="auto-indent" parentheses="no" brackets="no" curlyBrackets="no" quotes="no" doubleQuotes="no" htmlXmlTags="no" />
</GUIConfig>
```

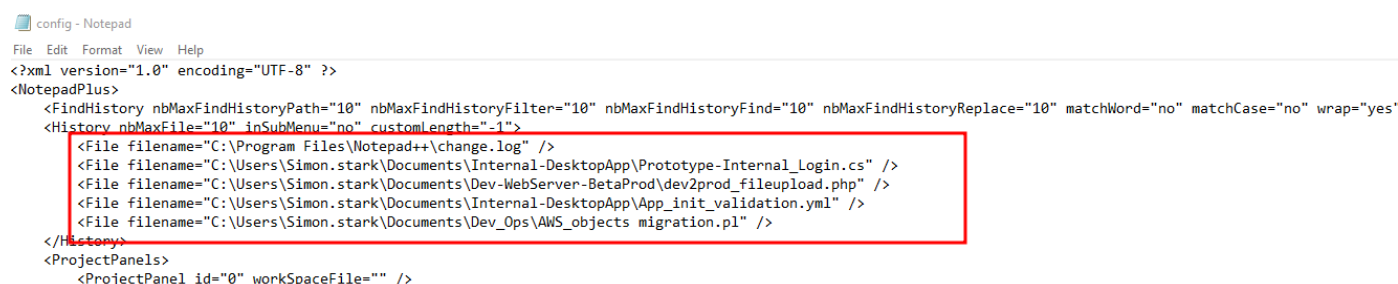
The `session.xml` file stores information about recent sessions. For example, if a document was opened in Notepad++ and the computer shut down or the application closed unexpectedly, the session file holds the necessary information to resume from where it was left off.

```
session - Notepad
File Edit Format View Help
<?xml version="1.0" encoding="UTF-8" ?>
<NotepadPlus>
  <Session activeView="0">
    <mainView activeIndex="1">
      <File firstVisibleLine="21" xOffset="0" scrollWidth="848" startPos="1697" endPos="1697" selMode="0" offset="0" wrapCount="1" lang="Java" encoding="-1" userReadOnly="no" />
      <File firstVisibleLine="0" xOffset="0" scrollWidth="1072" startPos="672" endPos="672" selMode="0" offset="0" wrapCount="1" lang="None (Normal Text)" encoding="-1" userReadOnly="no" />
    </mainView>
    <subView activeIndex="0" />
  </Session>
</NotepadPlus>
```

The backup folder stores unsaved data of files opened using Notepad++. If a document was being edited and the user forgot to save the file or the application crashed, that data would be stored in this folder to prevent data loss. Let's start answering the questions.

Q1 What is the full path of the script used by simon for AWS operations?

We will analyze the `config.xml` file, as it stores a history of the files opened by the application.



```
config - Notepad
File Edit Format View Help
<?xml version="1.0" encoding="UTF-8" ?>
<NotepadPlus>
  <FindHistory nbMaxFindHistoryPath="10" nbMaxFindHistoryFilter="10" nbMaxFindHistoryFind="10" nbMaxFindHistoryReplace="10" matchWord="no" matchCase="no" wrap="yes"
  <History nbMaxFile="10" inSubMenu="no" customLength="-1">
    <File filename="C:\Program Files\Notepad++\change.log" />
    <File filename="C:\Users\Simon.stark\Documents\Internal-DesktopApp\Prototype-Internal_Login.cs" />
    <File filename="C:\Users\Simon.stark\Documents\Dev-WebServer-BetaProd\dev2prod_fileupload.php" />
    <File filename="C:\Users\Simon.stark\Documents\Internal-DesktopApp\App_init_validation.yml" />
    <File filename="C:\Users\Simon.stark\Documents\Dev_Ops\AWS_objects migration.pl" />
  </History>
  <ProjectPanels>
    <ProjectPanel id="0" workspaceFile="" />
  </ProjectPanels>
</NotepadPlus>
```

We spot a few file paths. The last file, "AWS_objects migration.pl," is a Perl script that looks like it was written for some AWS-related tasks.

Answer: C:\Users\Simon.stark\Documents\Dev_Ops\AWS_objects migration.pl

Q2 The attacker copied an program code and compiled on the system as they knew the victim was a software engineer and had all those utilities. They did this in order to blend in the environment, and they didnt bring any of their own tools.This code collected sensitive data and made it ready to be exfiltrated. What's the full path of the program source file?

We can look at the `config.xml` file for historical files, but none of them seem malicious. Let's explore the backups directory.

Name	Date modified	Type	Size
 LootAndPurge	7/24/2023 3:05 PM	JAVA@2023-07-24...	3 KB
 YOU HAVE BEEN HACKED	7/24/2023 3:09 PM	TXT@2023-07-24_...	1 KB

In the backups directory, we find two cached files, both looking suspicious just by their names. Let's open the `LootAndPurge.java` file, as the extension indicates it's a Java file.

```

java
import java.io.File;
import java.io.FileInputStream;
import java.io.FileOutputStream;
import java.io.IOException;
import java.util.ArrayList;
import java.util.Arrays;
import java.util.List;
import java.util.zip.ZipEntry;
import java.util.zip.ZipOutputStream;

public class Sensitive_data_extort {
    public static void main(String[] args) {
        String username = System.getProperty("user.name");
        String desktopDirectory = "C:\\Users\\" + username + "\\Desktop\\";
        List<String> extensions = Arrays.asList("zip", "docx", "ppt", "xls", "md", "txt", "pdf");
        List<File> collectedFiles = new ArrayList<>();

        collectFiles(new File(desktopDirectory), extensions, collectedFiles);

        String zipFilePath = desktopDirectory + "Forela-Dev-Data.zip";
        String password = "sdklY57BLghvyh5FJ#fion_7";

        createZipArchive(collectedFiles, zipFilePath, password);

        System.out.println("Zip archive created successfully at: " + zipFilePath);
    }

    private static void collectFiles(File directory, List<String> extensions, List<File> collectedFiles) {
        File[] files = directory.listFiles();
        if (files != null) {
            for (File file : files) {
                if (file.isDirectory()) {
                    collectFiles(file, extensions, collectedFiles);
                } else {
                    String fileExtension = getFileExtension(file.getName());
                    if (extensions.contains(fileExtension)) {
                        collectedFiles.add(file);
                    }
                }
            }
        }
    }
}

```

We immediately see interesting function names. By reading the code, it seems this function gathers interesting data and prepares it for exfiltration. Given the scenario's context, we can confirm this file was used for data extortion.

To get the full path of the file, we will need to analyze the `session.xml` file. This file should have been populated with data related to the backup files.

```

:filename="C:\Users\Simon.stark\Desktop\LootAndPurge.java" backupFilePath="C:\Users\Simon.stark\AppData\Roaming\Notepad++\backup\LootAndPurge.java@2023-07

```

In the `session.xml` file, we find the path of the original file and the backup file we analyzed.

Answer: C:\Users\Simon.stark\Desktop\LootAndPurge.java

Q3 What's the name of the final archive file containing all the data to be exfiled?

From the previous question, we already saw the data extortion function, which collected all the sensitive data. We see the zip file name is hardcoded in the code, which may indicate this is a targeted extortion case.

```

String zipFilePath = desktopDirectory + "Forela-Dev-Data.zip";

```

Answer: Forela-Dev-Data.zip

Q4 What's the timestamp when attacker last modified the program source file? Please answer in UTC/GMT

We know that the source file in question is `LootAndPurge.java`. Now we need a timestamp to further enhance our incident response and help the SOC team build a timeline.

Let's analyze the `session.xml` file again, as it also stores the timestamps.

```
.ootAndPurge.java@2023-07-24_145332" originalFileLastModifTimestamp="-1354503710" originalFileLastModifTimestampHigh="31047188"  
originalFileLastModifTimestamp="-1354503710"  
originalFileLastModifTimestampHigh="31047188"
```

These values are in Windows FILETIME format. To convert these to UTC, we need to use a specific formula:

FILETIME timestamp=high×constant+(constant+low)
$$\text{FILETIME timestamp} = \text{high} \times \text{constant} + (\text{constant} + \text{low})$$

where the constant is $232 = 4294967296 \times 2^{32} = 4294967296 \times 232 = 4294967296$.

Inserting the values:

$$\text{FILETIME timestamp} = 31047188 \times 4294967296 + (4294967296 + (-1354503710))$$

$$\text{timestamp} = 31047188 \times 4294967296 + (4294967296 + (-1354503710))$$

I will use [this online calculator](#). Simply paste the above equation there.

Expression:

294967296 + (4294967296 - 1354503710)

Calculate

Result:

133346660033227230

he FILETIME timestamp is 133346660033227230. Converting this using an online FILETIME/LDAP timestamp converter. See the link for a converter - <https://www.epochconverter.com/ldap>.

Enter number in full or in scientific/exponential notation:

Milliseconds are discarded (last 7 digits of the LDAP timestamp)

133346660033227230

Convert 18-digit LDAP to human date/epoch

Epoch/Unix time: 1690192403

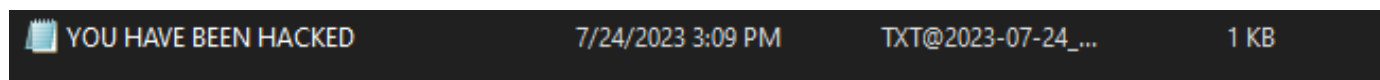
GMT: Monday, July 24, 2023 9:53:23 AM

Your time zone: Monday, July 24, 2023 2:53:23 PM GMT+05:00

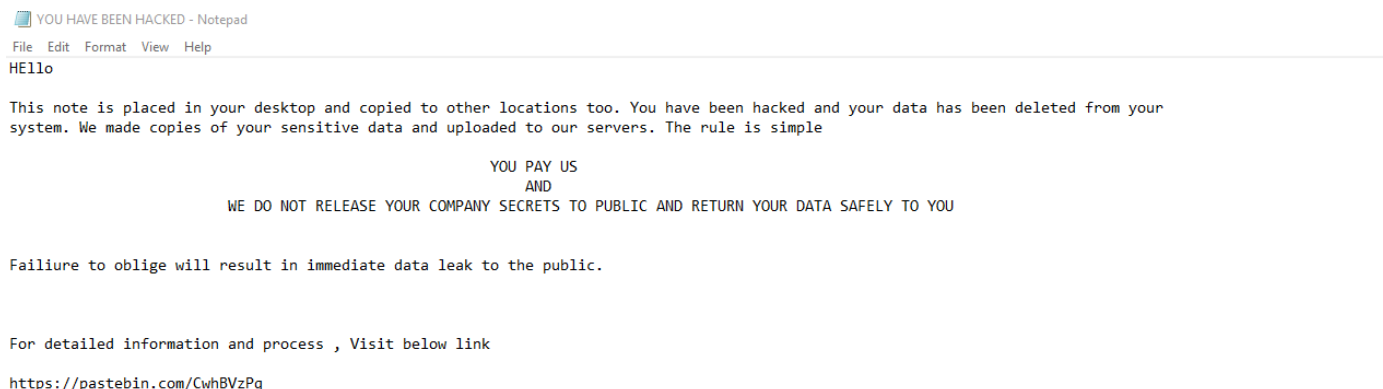
Answer: 07/24/2023 9:53:23

Q5 The attacker wrote a data extortion note after exfiltrating data. What is the crypto wallet address to which attackers demanded payment?

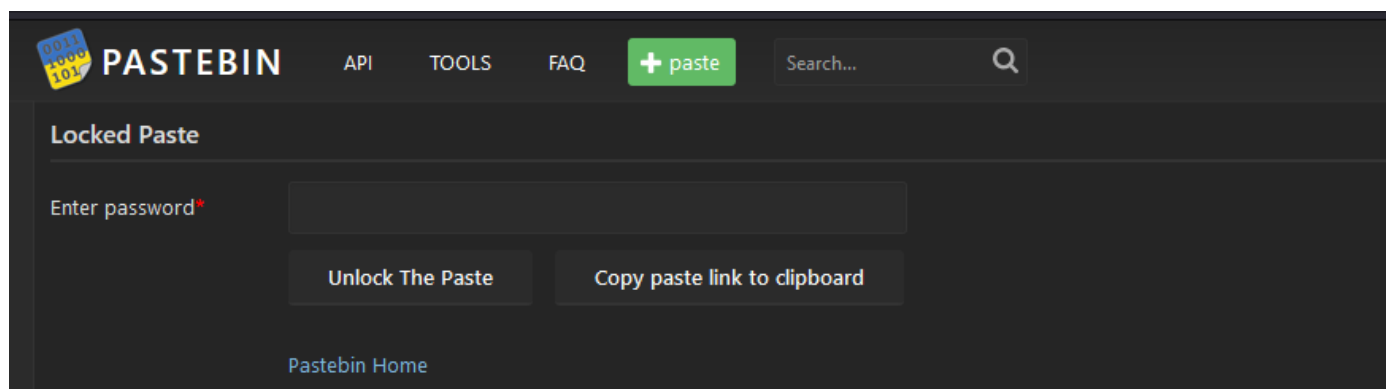
Let's return to the backup folder. We saw a txt file other than the Java file.



It looks like the data extortion note placed on the victim's computer.



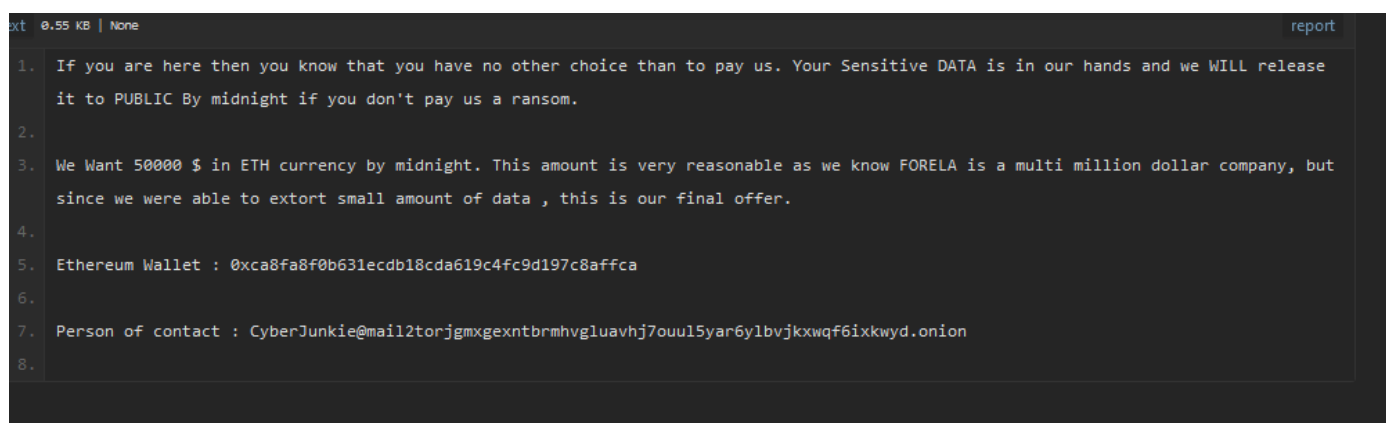
We see threats made by the attackers and a pastebin link. Visiting the link:



It is password locked, and the attacker didn't provide a password. The threat intelligence team believes the attackers accidentally locked the pastebin link. However, there was a hardcoded password in the Java source file.

```
String zipFilePath = desktopDirectory + "Forela-Dev-U  
String password = "sdk1Y57BLghvyh5FJ#fion_7";
```

Using this password on the pastebin link:



We get the full details provided by the threat actors, including the crypto wallet address.

Answer: 0xca8fa8f0b631ecdb18cda619c4fc9d197c8affca

Q6 What's the email address of person of contact for support?

We also get the email address of Person of contact to whom we can negotiate with.

Answer: CyberJunkie@mail2torjgmxgexntbrmhvgluavhj7ouul5yar6ylbvjkxwqf6ixkwyd.onion